The two-player $3n \pm 1$ game (Althöfer Prize 3): a retrograde analysis and a limitative result

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Abstract

I study the two-player impartial game on odd integers in which a move sends n to odd_part$(3n+1)$ or odd_part$(3n-1)$ and the player who moves to 1 wins (Althöfer Prize 3). I do **not** solve the game. I report two things I can back with exact computation. First, a Zermelo retrograde analysis over all odd $n \le 5 \times 10^5$: 73.0% of positions are resolved, split 61.0% WIN / 39.0% LOSE, with maximal optimal depth 79, mean 32.7, and — as it must be — every WIN position has at least one LOSE child (0 exceptions). Second, a *limitative* result: the WIN/LOSE label carries **no 2-adic residue structure**. Among the residue classes modulo 2^j there is not a single "pure" class (all-WIN or all-LOSE) for any $j \le 12$ — 0 of 4096 at $j = 12$. This is in sharp contrast with the *deterministic* accelerated Collatz map, whose descent behaviour *does* split into residue classes (I have 197 such classes machine-certified in Lean 4). The consequence is concrete: no ranking function that reads only a bounded number of low bits of n can decide this game.

1 The game

The game is Prize 3 of Althöfer's list of Collatz prizes [1]; for a closely related two-player Collatz game and other variants of the $3n+1$ problem see [3]. On the odd integers, a position $n > 1$ has two moves,

$$n \longmapsto \text{odd_part}(3n+1) \qquad \text{or} \qquad n \longmapsto \text{odd_part}(3n-1),$$

where odd_part$(x) = x/2^{v_2(x)}$. The position 1 is terminal; the player who moves to 1 wins. The game is *impartial* (both players share the same two moves from any position), so by Zermelo's theorem [4] every position is either an N-position (the player to move wins under optimal play; "WIN") or a P-position ("LOSE"), computed by backward induction from 1:

- 1 is a P-position (LOSE for the player to move);
- a position with *some* move to a P-position is N (WIN);
- a position *all* of whose moves lead to N-positions is P (LOSE).

Observation 1 (At least one move shrinks)**.** *For every odd $n > 1$, exactly one of $3n+1, 3n-1$ is $\equiv 2 \pmod 4$ and the other $\equiv 0 \pmod 4$. Hence one move goes "up" by a factor $\approx \frac{3}{2}$ and the other "down" by a factor $3/2^k$ with $k \ge 2$: at least one legal move strictly decreases n. So every orbit under free play can be driven towards 1, and the game is well-founded.*

2 Retrograde analysis

I ran the exact backward induction over all odd $n \le N = 5 \times 10^5$. Moves may leave the box $[1, N]$; a position whose analysis needs a value outside the box stays unresolved. This is the coverage figure, not a defect of the method.

odd positions $n \le 5 \times 10^5$	250,000
resolved	182,559 (73.0%)
WIN (N)	111,378 (61.0% of resolved)
LOSE (P)	71,181 (39.0%)
maximal optimal depth (over WIN)	79
mean optimal depth (over WIN)	32.7
WIN positions with no LOSE child	0

Table 1: Zermelo retrograde analysis of the $3n \pm 1$ game.

The last row is the structural sanity check: a WIN position *must* have a move to a LOSE position, and it does, in all 111,378 cases. The depth is the number of moves to the end under optimal play (the winner finishes fast, the loser stalls); depth 79 is the longest optimal line I found.

Remark 2 (On modelling)**.** This is a *two-player* game; the relevant quantity is the Zermelo N/P label and the optimal depth, not the length of a shortest path to 1. A shortest-path (solitaire) reading gives a different, smaller depth distribution and answers a different question. The numbers above are from the two-player retrograde analysis.

3 The limitative result: no 2-adic structure

The natural first attempt at Prize 3 is to look for a *ranking*: a function $\rho(n)$ that some player can force to decrease, or equivalently a description of the N/P label by simple arithmetic of n. For the *deterministic* accelerated Collatz map this strategy works in part — the descent behaviour splits into residue classes modulo powers of 2 (e.g. $n \equiv 1 \pmod 4$ descends in one step), and I have 197 such minimal descent classes (1399 theorems) machine-certified in Lean 4. One might hope the game label is similarly 2-adic.

Proposition 3 (No pure residue class)**.** *For every $j \le 12$,* not one *of the 2^j residue classes modulo 2^j is pure: each class that occurs contains both WIN and LOSE positions. In particular at $j = 12$,* 0 *of the* 4096 *classes are pure.*

This is an exhaustive computation over the resolved positions $n \le 5 \times 10^5$; the LOSE positions are moreover equidistributed modulo 3, 4 and 8. The contrast with the deterministic map is the whole point:

	deterministic Collatz descent	the $3n \pm 1$ *game* label
residue classes mod 2^j	split cleanly (197 certified)	**0 pure**, every $j \le 12$

Proposition 4 (Consequence)**.** *No ranking function that depends only on $n \bmod 2^j$ (for any fixed $j \le 12$) can decide the WIN/LOSE label of the game. Any residue-based strategy of bounded 2-adic depth is therefore ruled out.*

Proof. If the label were a function of $n \bmod 2^j$, every class modulo 2^j would be pure. By Proposition 3 none is, for any $j \le 12$. □

The only ranking I found that does decrease is the game depth $d(n)$ itself — but that is circular, since d is *defined* by solving the game. Every non-circular residue ranking I tried (by n, by bit-length, by popcount, by $\log_2 n + r(n)$) fails on a positive fraction of positions.

4 What I claim, and what I do not

- I do **not** solve Prize 3. The game is open.
- The retrograde numbers (73.0%, 79, 32.7, structural check) are exact over $n \leq 5 \times 10^5$.
- Proposition 3 is an exhaustive check to $j \leq 12$; I state it with that range, not as "for all j". It is strong evidence that the game has no finite 2-adic structure, and it closes off the residue-ranking approach up to that depth.
- Observation 1 (at least one move shrinks) is proved.

I would value your view on whether the absence of 2-adic structure is the expected state of affairs for this game, or whether it points to where a solution must *not* be looked for.

Reproducibility

All numbers are regenerated by a single self-contained script (`verify_prize3.py`); the 197 certified Collatz descent classes build in Lean 4 with `lake build` and pass `#print axioms` with kernel-only axioms.

Acknowledgement

This work is AI-assisted; every number reported here was regenerated by exact computation before inclusion, and the modelling was corrected (two-player retrograde analysis, not shortest path) after an initial mistake.

References

[1] I. Althöfer, *Collatz prizes*, https://althofer.de/collatz-prizes.html.

[2] J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, American Mathematical Society, 2010.

[3] I. Althöfer, M. Hartisch and T. Zipproth, *Analysis of a Collatz game and other variants of the* $3n+1$ *problem*, in: Advances in Computer Games, LNCS, Springer, 2024, 123–132. doi:10.1007/978-3-031-54968-7_11.

[4] E. Zermelo, *Über eine Anwendung der Mengenlehre auf die Theorie des Schachspiels*, in: Proc. Fifth Int. Congress of Mathematicians (Cambridge, 1912), vol. II, 1913, 501–504. JFM 44.0092.04.